

supply chains, recognising that climate change is no longer a mere planetary imperative but a business continuity imperative.

Breakthroughs in new technology around green hydrogen, autonomous mobility, bio-engineered seed varieties and more are already emerging. These will be necessary to aid decarbonisation of critical sectors of the economy—from energy to mobility and supply chains to agriculture.

With 70 per cent of energy still being fossil-fuel-based, it contributes 40 per cent to India's emissions. Energy usage in the country has doubled over the past decade and it is expected to grow at three times the global average. Transition to cleaner power production will not only help reduce GHG emissions, but also enable energy security. India already has a target of 50 per cent renewable capacity installation by 2030.

Efficient monitoring, management and optimised use of natural resources such as air, water, land and biodiversity will be important to prevent further emissions and environmental degradation, and

they can help preserve and replenish critical resources. As our economy grows, we are also projected to be the largest waste contributor of the world by 2048, generating 125 million tonnes of waste. At present, only 20 per cent of this waste is recycled, necessitating circularity and recycling for end-of-life products.

There is significant policy push towards achieving electrification of transport with nearly 80 per cent EV penetration envisioned by 2030 in two-wheelers and three-wheelers that constitute a majority of road traffic in India. This, combined with progressive greening of the grid, will be exponential. Current roadblocks on poor charging infrastructure networks, unavailability of finance, etc. are progressively being solved and will enable faster pace of deployment.

India's supply chain and industry is characterised by high fragmentation, inefficiencies and low digitisation as our supply chain costs are at 14 per cent of GDP against an average of 8 per cent globally. Digitalisation of supply chains is taking place through both horizontal and vertical solutions. Technology-enabled

sourcing, production, distribution and end-of-life processes will reduce wastage. Similarly, alternative low-GHG materials in industries like construction, consumer goods and industrial goods will significantly reduce emissions.

Agriculture and food security will be vital cogs in this transition, especially given that 50 per cent of our population depends on agriculture for their livelihood. The sector is also a large resource drain, accounting for 80 per cent of water usage and nearly 60 per cent of our land usage. As our population increases, food production needs to be doubled by 2050. Tech-led solutions such as precision agriculture and the use of IoT are enabling optimisation of resource usage for food production that can reduce its environmental footprint. Alternative climate-resilient and resource-efficient farming methods such as controlled-environment agriculture will solve for food security challenges.

It is evident that investment in cutting-edge technology and innovation will be required to create low-carbon and climate-resilient development pathways that can unlock outcomes for social, economic and environmental impact by empowering underserved segments, unlocking inclusive growth and enabling sustainability. Moreover, solutions created from India will have global applications—not just in similar developing economies in the Middle-east, South-East Asia and Africa, but also for developed countries in the global North. India has claimed its place on the international stage, and is well-positioned to benefit from geopolitical tailwinds to establish itself as a pioneer in global climate action and sustainability. **BT**

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